

# EXP-010 — Encoder-Window-Aligned Chunking

Production RAG v1 · evaluation-first · n = 20 questions / 22 evidence spans

STATUS — HYPOTHESIS FALSIFIED

Truncation was eliminated completely — **23.22% of chunks truncated** → **0%**, corpus token coverage **0.7610** → **1.0000** — and retrieval did not move at all: **Δ0.000**, **zero cases rescued**, **zero regressed**, on both primary comparisons. Encoder visibility correlated with the EXP-009 result but is **not the causal bottleneck**.

+0.000

B → D transformer  
control → aligned

+0.000

C → E fusion  
mixed representation

1 / 22

answers actually hidden  
by the 512 window

1.00

corpus token coverage  
(control 0.76)

## 1. Why EXP-010 exists

EXP-009 measured the same transformer at two windows and found retrieval tracked how much of a unit the encoder could see (256 tokens: 35.2% truncated, recall 0.500; 512 tokens: 23.2% truncated, recall 0.575, fused 0.775). That was a correlation across two settings with one observation each. EXP-010 holds the window fixed at 512 and changes the retrieval unit instead, so the claim can be tested rather than repeated.

## 2. Not another "smaller chunks" experiment

EXP-005 (chunk size × BM25) and EXP-008 (chunk size × dense) both shortened chunks using character heuristics and both failed; EXP-008 showed splitting a coherent unit makes dense retrieval *worse*. Here every limit is measured in the encoder's own WordPiece tokenization, and the objective is the **largest coherent unit the encoder can consume whole** — explicitly not the smallest.

The chunk set is **derived from the control**: a chunk that already fits passes through with the same source span and text, and only an oversized chunk is split. That decision was committed before the build, so D vs B isolates encoder alignment rather than confounding it with re-grouping.

## 3. Encoder budget — measured, not assumed

| quantity                | value     | source                                   |
|-------------------------|-----------|------------------------------------------|
| max_position_embeddings | 512       | config.json                              |
| special-token overhead  | 2         | probe encoded with and without specials  |
| usable payload          | 510       | 512 – 2                                  |
| target / hard payload   | 448 / 480 | conservative, near the top of the window |

## 4. Intervention fidelity

|                          | control | encoder-aligned |
|--------------------------|---------|-----------------|
| chunks                   | 14,209  | 18,579          |
| median encoded tokens    | 146     | 181             |
| p95                      | 881     | 445             |
| max                      | 6,857   | 482             |
| chunks over 512          | 3,300   | 0               |
| corpus token coverage    | 0.7610  | 1.0000          |
| evidence spans preserved | 22/22   | 22/22           |

Verified twice independently: the gate script tokenized all 18,579 chunks, and the embedding build measured truncation through the encoder's own path while encoding ( `texts_truncated: 0` , `token_coverage: 1.0` ). The corrected control figures match EXP-009's independently recorded numbers exactly.

## 5. Reproduction gates

| cell                                | checks                                                                   | verdict |
|-------------------------------------|--------------------------------------------------------------------------|---------|
| A — BM25, control chunks            | macro_span_recall 0.475 · cases_fully_recalled 9 · spans_found_at_10 10  | PASS    |
| B — transformer @512, control       | macro_span_recall 0.575 · cases_fully_recalled 11 · spans_found_at_10 13 | PASS    |
| C — BM25 + transformer RRF, control | macro_span_recall 0.775 · cases_fully_recalled 15 · spans_found_at_10 17 | PASS    |

## 6. Results

| cell                                   | macro R | full  | spans@10 | doc R | MRR   | a@10 | a@50 | a@300 | ms  |
|----------------------------------------|---------|-------|----------|-------|-------|------|------|-------|-----|
| A — BM25, control chunks               | 0.475   | 9/20  | 10/22    | 0.825 | 0.280 | 12   | 3    | 1     | 421 |
| B — transformer @512, control          | 0.575   | 11/20 | 13/22    | 0.925 | 0.339 | 9    | 4    | 1     | 199 |
| C — BM25 + transformer RRF, control    | 0.775   | 15/20 | 17/22    | 0.925 | 0.449 | 5    | 3    | 2     | 575 |
| D — transformer @512, encoder-aligned  | 0.575   | 11/20 | 13/22    | 0.925 | 0.338 | 9    | 5    | 1     | 190 |
| E — BM25 control + transformer aligned | 0.775   | 15/20 | 17/22    | 0.875 | 0.460 | 5    | 3    | 2     | 585 |

B and D are identical on every headline metric. C and E are identical on recall; E's MRR is marginally better (0.460 vs 0.449) and its document recall marginally worse — both well inside noise at n = 20.

## 7. Why the null is not "nothing happened"

**12 of 22 spans changed rank.** The movements simply never crossed the top-10 boundary. Splitting them by cause is what makes the result interpretable:

- **7 spans** sat in a truncated chunk, so their carrying chunk genuinely changed — and their cosine changed with it. 3 improved, 4 worsened.
- **5 spans** sat in chunks that already fitted and passed through byte-identical — their cosine is *identical to six decimals*. They moved only because other documents' chunks were resplit and the competition changed.

Rank is relative: an untouched chunk still moves when the corpus around it is recut. It also means the EXP-008 fragmentation failure did **not** repeat — units that already fitted were never touched.

## 8. The decisive measurement: chunk truncation is not answer invisibility

| case          | chunk tokens | answer offset | visible at 512? | rank B → D | movement |
|---------------|--------------|---------------|-----------------|------------|----------|
| AN-002        | 848          | 346           | yes             | 18 → 26    | worsened |
| AN-003        | 801          | 118           | yes             | 299 → 193  | improved |
| AN-004        | 796          | 235           | yes             | 2 → 3      | worsened |
| <b>AN-007</b> | 848          | 588           | <b>no</b>       | 16 → 28    | worsened |
| AN-010        | 671          | 397           | yes             | 49 → 127   | worsened |
| AN-012        | 848          | 407           | yes             | 8 → 6      | improved |
| OA-002        | 620          | 102           | yes             | 7 → 2      | improved |

### THE FINDING THAT EXPLAINS THE NULL

**Only one answer out of 22 — AN-007 — was actually outside the visible window.** The other 21 were already fully visible to the encoder. 23% of *chunks* were truncated, but only 4.5% of *answers* were hidden, because answers sit near the top of the sections that carry them.

There was therefore almost nothing left for encoder alignment to fix — and the one case where the hypothesis made a direct prediction, **AN-007, got worse when its answer was made visible** (16 → 28). The single span the mechanism was supposed to rescue moved the wrong way.

## 9. AN-003, AN-002 / AN-007, OA-004

**AN-003** improved 299 → 193 with a rising cosine — the largest single movement in the experiment, and still far outside any practical pool. Its answer was *already visible* at token offset 118, so this is not a truncation rescue. AN-003 has now failed under BM25, FastText, the transformer at two windows, and encoder-aligned chunking.

**AN-002** (18 → 26) and **AN-007** (16 → 28) both worsened; neither was inside the top 10 to begin with. **OA-004** moved 24 → 26 under D and is unchanged between the fused cells, so the EXP-009 fusion rescue is not undone. It is still unsolved.

## 10. Reranker decision gate — counts only, no reranker was built

| cell                                   | 1-10 | 11-30 | 31-50 | 51-100 | 101-300 | absent | ceil@30 | ceil@50 | ceil@100 |
|----------------------------------------|------|-------|-------|--------|---------|--------|---------|---------|----------|
| A — BM25, control chunks               | 10   | 7     | 2     | 1      | 1       | 1      | 0.773   | 0.864   | 0.909    |
| B — transformer @512, control          | 13   | 4     | 1     | 0      | 3       | 1      | 0.773   | 0.818   | 0.818    |
| C — BM25 + transformer RRF, control    | 17   | 1     | 1     | 1      | 0       | 2      | 0.818   | 0.864   | 0.909    |
| D — transformer @512, encoder-aligned  | 13   | 4     | 0     | 0      | 4       | 1      | 0.773   | 0.773   | 0.773    |
| E — BM25 control + transformer aligned | 17   | 1     | 1     | 1      | 0       | 2      | 0.818   | 0.864   | 0.909    |

Alignment did not improve rerankable headroom; at pools 50 and 100 it slightly **reduced** it. A reranker would work against a ceiling of 0.909 with the fused retriever already delivering 0.775 — roughly three spans of headroom across 20 questions. Not yet a compelling case.

## 11. EXP-NULL

**BLOCKED.** `api.anthropic.com` is reachable but answers `401` without a key; `api.openai.com` remains blocked at the egress proxy. No credential was fabricated, printed, or inferred, and the session's own harness credentials are not a project credential. EXP-010 was not delayed for it.

## 12. Limitations

- $n = 20 / 22$  spans. One case is 5 percentage points. No significance is claimed.
- A  $\Delta 0.000$  with 12 spans moving is a *null at the decision boundary*, not proof of no effect. A different top-k or a larger question set could separate B and D.
- The answer-offset measure counts tokens to the *start* of the evidence span; an answer beginning inside the window but extending past it counts as visible.
- One encoder, one window (512, inherited from EXP-009 and deliberately not swept).

## 13. Did encoder alignment earn promotion?

**No.** The frozen production baseline stays control chunks / no enrichment / BM25 / `top_k=10`. The strongest measured configuration remains **BM25 + transformer @512 RRF on control chunks — 0.775 / 15-of-20**. The aligned set costs 31% more chunks, 48 MB more vectors and a longer build, and returns nothing.

## 14. What the measurements justify next

1. **Stop treating truncation as the bottleneck.** It is fixed completely and bought nothing. Three chunking interventions have now returned 0, -0.050 and 0.000 — **chunking is not where the remaining recall is.**

2. **The gap is ranking, not visibility.** 21 of 22 answers were already visible and 9 still sit outside the top 10. The encoder sees the right text and scores it below other text.
3. **Investigate the query side, which has never been touched.** Every experiment so far changed the document representation; the query has been a raw question string since EXP-000.
4. **A reranker is still not justified** — ceiling 0.909 against 0.775 delivered.
5. **AN-003 needs a failure report, not another chunker.**

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Generated from experiments/EXP-010/\*.json by scripts/build\_exp010\_pdf.py. Git commit 34dacc4d57e9. Config hash dd050bde901cb3a6. Encoder sentence-transformers/all-MiniLM-L6-v2 @ 512 tokens, fingerprint bd95feaeacf98559, exact cosine, no ANN index. Chunker chunker\_v4\_encoder\_aligned v4.0. Raw provider documentation is not redistributed.